

Beyond Static Imitation: Preserving LLM-Derived Traveler Responses for Transport Simulation

An Xu^a, Zekai Jin^b and Yunfei Yin^{a,*}

^a*School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin 150090, China*

^b*Department of Civil Engineering, McGill University, Montreal, QC H3A 0C3, Canada*

ARTICLE INFO

Keywords:

behavioral distillation
response fidelity
large language models
stated choice
agent-based transport simulation

ABSTRACT

Large language models (LLMs) can describe how travelers might react to delays, fares or rain, but querying them for every agent in a city-scale simulation remains costly. Distilling their judgments into compact traveler models removes this cost and raises a question that predictive accuracy does not answer, namely whether the distilled model changes its choices in the same way as its source when conditions change. This study evaluates LLM-derived traveler models by their responses to paired changes in travel conditions and follows these responses through comparisons with the LLM Teacher, with stated choices and with simulated network demand. On a held-out benchmark of six synthetic personas and three training seeds, paired response supervision lowers probability-response error by 4.97% relative to pointwise distribution matching, and a multinomial logit Student lowers it by a further 14.30%. Combining that choice model with an independent neural departure predictor gives a departure error of 7.61 minutes against the Teacher, within the range of jointly trained neural models. Stated-choice data from 332 Singapore and 321 Shanghai respondents show that agreement with people depends on the intervention, and the model with the smallest Teacher error reverses the stated response to poorer public-transport accessibility. Contemporary Teacher predictions for 24 respondents per city distinguish Teacher-human from Student-Teacher discrepancies on the same tasks. In a 1,000-traveler Helsinki simulation, feasibility-constrained assignment makes more trips routable without narrowing the gap between predicted and simulated responses. Fidelity at one level therefore need not carry over to the next, and the choice of traveler model should follow the intervention and behavioral outcome that a simulation is meant to inform.

1. Introduction

Transport scenario analysis is concerned less with where travelers go today than with how their choices would change if conditions changed. A public-transport delay may push some commuters into a car, leave others on the same service and lead others to depart earlier. Rain, higher fares and longer walks to a stop produce different reactions, even among people making similar trips. Agent-based simulation represents how such decisions interact with road and transit supply (Horni et al., 2016; Axhausen, 2002), and the policy value of a simulated scenario therefore rests on the change in demand that its behavioral model generates, not only on the baseline mode shares it reproduces.

Large language models (LLMs) have become an attractive source for this behavioral component. A single prompt can combine a traveler's circumstances, the available modes and the attributes of a trip with a change in context, and recent studies have used this flexibility for travel-behavior prediction, route-choice agents and agent-based transport modeling (Mo et al., 2026; Wang et al., 2025; Liu et al., 2025; Nie et al., 2025). Two obstacles stand between these demonstrations and city-scale use. Querying an LLM for every agent and scenario is slow and costly, and it remains unclear how closely LLM judgments track the behavior of actual travelers. Liu et al. (2025) identify both issues as open problems for LLM-agent modeling and suggest hybrid designs that combine LLM agents with conventional models as a near-term route. Behavioral distillation is one such design. The LLM acts as a Teacher that supplies judgments offline, and a compact numerical Student makes the repeated decisions that a simulation requires.

Distillation shifts attention to a question that predictive accuracy does not settle, namely which properties of the Teacher the Student must retain. Evidence from outside distillation suggests that the answer is not obvious. LLMs used as synthetic survey respondents can reproduce aggregate patterns of human answers (Argyle et al., 2023; Aher et al., 2023), yet their answers may show too little variation and depend on prompt wording and model version (Bisbee

*Corresponding author: Yunfei Yin (yunfeiyin@hit.edu.cn).

Email addresses: AnX.RTL2324@tum-asia.edu.sg (A. Xu); yunfeiyin@hit.edu.cn (Y. Yin)

et al., 2024). In travel choice experiments, LLMs recover the direction of value-of-time effects more reliably than their sensitivity to context and income (Yan et al., 2026). Machine-learning choice models raise a parallel concern, because high predictive accuracy does not guarantee plausible elasticities, substitution patterns or other behavioral quantities (Wang et al., 2020; Martín-Baos et al., 2023). In each case the property most relevant to policy, sensitivity to a change in conditions, is also the property that accuracy measures least directly.

Two targets of distillation can therefore be distinguished. *Static fidelity* refers to agreement between Student and Teacher at individual traveler states. *Response fidelity* refers to agreement in the change of mode-choice probabilities between paired states that hold the traveler and trip fixed (Figure 1). Two models can share the same most likely mode at baseline and still redistribute probability very differently when a delay or fare increase is introduced. For scenario analysis, the second kind of agreement determines how a modeled intervention changes demand.



Figure 1: Traveler profiles, contextual changes and aggregate demand responses. Each scenario pair holds the traveler and trip fixed. The profiles are illustrative. The bars show mode assignments before routing for 10,000 synthetic Singapore travelers under the supply-adapted Student introduced in Section 3.2, rounded to whole percentages. Poorer accessibility is evaluated separately. Scenario definitions and exact shares are given in Supplementary Table S29.

Response fidelity to the Teacher is necessary for a useful Student but not sufficient. An LLM may hold preferences that differ from those of travelers, so a faithful Student can inherit the Teacher's biases along with its useful responses. Paired stated-choice tasks offer an external reference for these changes (Louviere et al., 2000). A predicted response must also pass through mode assignment and routing before it becomes simulated demand, and both steps can alter it while the behavioral model stays fixed. An LLM-derived traveler model is thus exposed to three distinct comparisons, with its Teacher, with people and with the simulated network. This study examines whether good performance in one of these comparisons carries over to the others.

The analysis combines a controlled Teacher-response benchmark, stated-choice experiments in Singapore and Shanghai with a contemporary Teacher comparison on the same tasks, and a paired MATSim experiment in Helsinki (Figure 2). Within the benchmark, learning objectives that weight paired probability changes explicitly are compared with conventional distribution matching, and a jointly trained neural Student is compared with a modular specification in which linear choice utilities and a departure-time predictor are estimated separately. The human and simulation experiments then follow these models beyond the Teacher.

Behavioral distillation is formulated here around finite responses to specified changes in travel conditions, and supervising these responses reduces Teacher-response error in a matched neural comparison. A larger gain comes from model specification than from the training objective. A multinomial logit Student distilled from the same targets preserves Teacher responses better than every neural Student, and pairing it with an independent neural departure predictor recovers timing accuracy comparable to that of the joint models. Beyond the Teacher, the stated-choice and simulation experiments show that neither Teacher fidelity nor route feasibility orders the models in the way that agreement with people or preservation in simulation does. These findings argue for judging LLM-derived traveler models against the intervention and behavioral outcome that a simulation is meant to inform, rather than against a single imitation score.

2. Related work

2.1. Language models as sources of travel behavior

Transport research has begun to place LLMs directly inside traveler agents. GATSim and related generative-agent frameworks use them to plan activities and travel in urban mobility simulation (Liu et al., 2026a,b), and LLM agents have been used to model day-to-day route choice (Wang et al., 2025). Other studies treat the LLM as a predictor, supplying retrieved trip examples for mode-choice prediction (Xu and Jiao, 2027) or supporting class-incremental learning as the prediction task evolves (Chen et al., 2026). Conceptual work in this journal has outlined how LLM agents could be organized within activity-based modeling and where their limitations lie (Liu et al., 2025; Nie et al., 2025). In all of these designs the language model is queried whenever a decision is made.

A second body of work asks how closely LLM outputs resemble human behavior. Studies of synthetic survey respondents report that LLMs can recover aggregate patterns and classic experimental effects (Argyle et al., 2023; Aher et al., 2023), while also documenting compressed variation and dependence on prompt wording and model version (Bisbee et al., 2024). Comparisons of intertemporal choice show systematic departures from human patience (Goli and Singh, 2024). For travel, Yan et al. (2026) find that LLM choices reproduce the direction of value-of-time effects more consistently than their magnitude across contexts, and Liu et al. (2026c) calibrate LLM agents to observed learning and adjustment behavior. These studies evaluate the LLM itself. When its judgments are compressed into a cheaper model, a further question arises about which of these properties survive, and whether a Student's discrepancies from people originate in the Teacher or in the compression.

2.2. Behavioral outputs of choice models

Random-utility models relate mode probabilities to traveler characteristics and alternative attributes (McFadden, 1974; Ben-Akiva and Lerman, 1985), and generalized and mixed logit formulations represent richer heterogeneity and correlation (Walker and Ben-Akiva, 2002; Hensher and Greene, 2003; Train, 2009). Neural networks can learn part of the utility function while keeping an interpretable core (Sifringer et al., 2020). Work comparing these families has repeatedly found that predictive accuracy and behavioral plausibility can diverge. Wang et al. (2020) show that elasticities and other economic quantities extracted from deep neural networks can be unstable even when predictive performance is high, and Martín-Baos et al. (2023) report that individual prediction, market shares and behavioral indicators need not favor the same model.

This literature fits models to observed choices. In distillation the Student fits another model, and its targets are soft probabilities and departure adjustments defined for arbitrary synthetic states. Whether flexible neural specifications or structured linear utilities better preserve such targets under controlled perturbations has received little attention. Mode choice and departure-time adjustment may also call for different functional forms, which motivates comparing a jointly trained neural Student with a modular alternative that estimates the two outcomes separately.

2.3. Distilling responses for simulation

Knowledge distillation transfers a Teacher's predictions or representations to a smaller Student (Hinton et al., 2015; Gou et al., 2021), and DistilBERT is a well-known example of compressing a language model (Sanh et al., 2019). Relational distillation extends the idea from individual outputs to relationships between examples (Park et al., 2019). The present setting is closer to the relational case. Its supervision consists of elicited mode probabilities and departure adjustments rather than generated text, and the relationship of interest is the change in these probabilities when the same traveler faces different conditions.

If a Student matched the Teacher exactly at every state, it would also match every such change. With limited training coverage and model capacity, endpoint errors do not cancel, and weighting the differences explicitly may improve the

contrasts on which scenario analysis depends. Whether a preserved contrast then reaches the simulated network is a separate matter. In MATSim, a predicted mode must be assigned to a person, routed on road and public-transport supply and executed alongside other plans (Horni et al., 2016). Evaluations of LLM-based traveler models have rarely followed a behavioral response through these steps, although it is at this point that the response becomes demand.

3. Behavioral response distillation

Behavioral distillation is treated here as the estimation of a surrogate that reproduces both the Teacher’s predictions at individual traveler states and its reaction to specified changes in travel conditions. Every candidate model produces a mode-choice distribution and a departure-time adjustment, which allows joint and modular specifications to be compared on the same behavioral outcomes. How these outcomes are realized in simulation is examined separately (Figure 2).

3.1. Traveler states and response targets

A state $\mathbf{x}$ describes a traveler, a trip and its context, together with level-of-service attributes for four alternatives: private car, public transport (PT), bicycle and walking. The attributes cover travel time, monetary cost and weather exposure, and for PT also access, waiting, in-vehicle and transfer conditions. Variable definitions and encoding rules are given in Supplementary Section S1 and Table S6.

Let $\mathbf{p}_j(\mathbf{x})$ denote mode-choice probabilities and $\tau_j(\mathbf{x})$ the departure adjustment in minutes, where $j \in \{T, S\}$ indexes the Teacher and the Student. For a pair of states that share the same traveler and trip,

$$\Delta \mathbf{p}_j(\mathbf{x}, \mathbf{x}') = \mathbf{p}_j(\mathbf{x}') - \mathbf{p}_j(\mathbf{x}), \quad j \in \{T, S\}. \quad (1)$$

These differences are the response targets. The perturbation from $\mathbf{x}$ to $\mathbf{x}'$ is a specified change to the state, such as additional PT delay or heavy rain. The resulting finite contrasts describe how a model reacts to that change. They are not, on their own, estimates of elasticities or of causal effects in a traveler population.

The Teacher targets are numerical judgments elicited from the DeepSeek model deepseek-v4-pro. They are neither observed choice frequencies nor token-level probabilities. When several valid probability vectors are available for a state, they are averaged, so that repetition reflects variability in the Teacher rather than additional travelers. The supervision used for training and the later Teacher evaluation are described separately in Supplementary Table S38.

3.2. Joint and modular behavioral specifications

The joint neural Student learns choice and timing from shared representations. A global encoder embeds traveler and trip attributes, an alternative encoder embeds mode-specific attributes, and a shared scorer maps both to a utility u_m . Probabilities are normalized over the alternatives available to the traveler, $\mathcal{A}(\mathbf{x})$:

$$p_{S,m}(\mathbf{x}) = \frac{\exp(u_m)}{\sum_{k \in \mathcal{A}(\mathbf{x})} \exp(u_k)}, \quad m \in \mathcal{A}(\mathbf{x}). \quad (2)$$

Unavailable modes receive zero probability. Departure adjustment is predicted from the global representation and the mean representation of the available alternatives, with $\tau_S = 60 \tanh(z)$ limiting the adjustment to ± 60 minutes. Network dimensions are reported in Supplementary Section S2.

The supply-matched multinomial logit Student (MNL-S) replaces the neural scorer with four linear utilities on the same structured information and is fitted to the soft Teacher targets. Choice and timing are estimated separately in this specification. Three timing predictors are paired with the same MNL-S choice probabilities: ridge regression, a bounded linear predictor $60 \tanh(\mathbf{w}^\top \mathbf{x} + b)$, and an independent neural predictor that shares the joint model’s encoder structure but receives only departure supervision. Holding the choice model fixed isolates the contribution of the timing specification.

A third model, the *supply-adapted Student* (SA-Student), extends a pretrained joint model with additional PT-supply attributes and with supervision on contrasts between travelers, between contexts and attributes, and between accessibility levels. The adapted model was selected by minimizing KL divergence on the accessibility validation states, subject to constraints on its earlier performance (Supplementary Table S25), and it is held fixed in the stated-choice and simulation experiments. Because its initialization, objective, auxiliary supervision and treatment of mode availability all differ from those of the controlled models, SA-Student is reported as an additional model rather than as a test of the response loss. The roles of the three model families are summarized in Supplementary Table S36, and the adaptation is specified in Section S4.

Response-aware behavioral distillation and evaluation framework

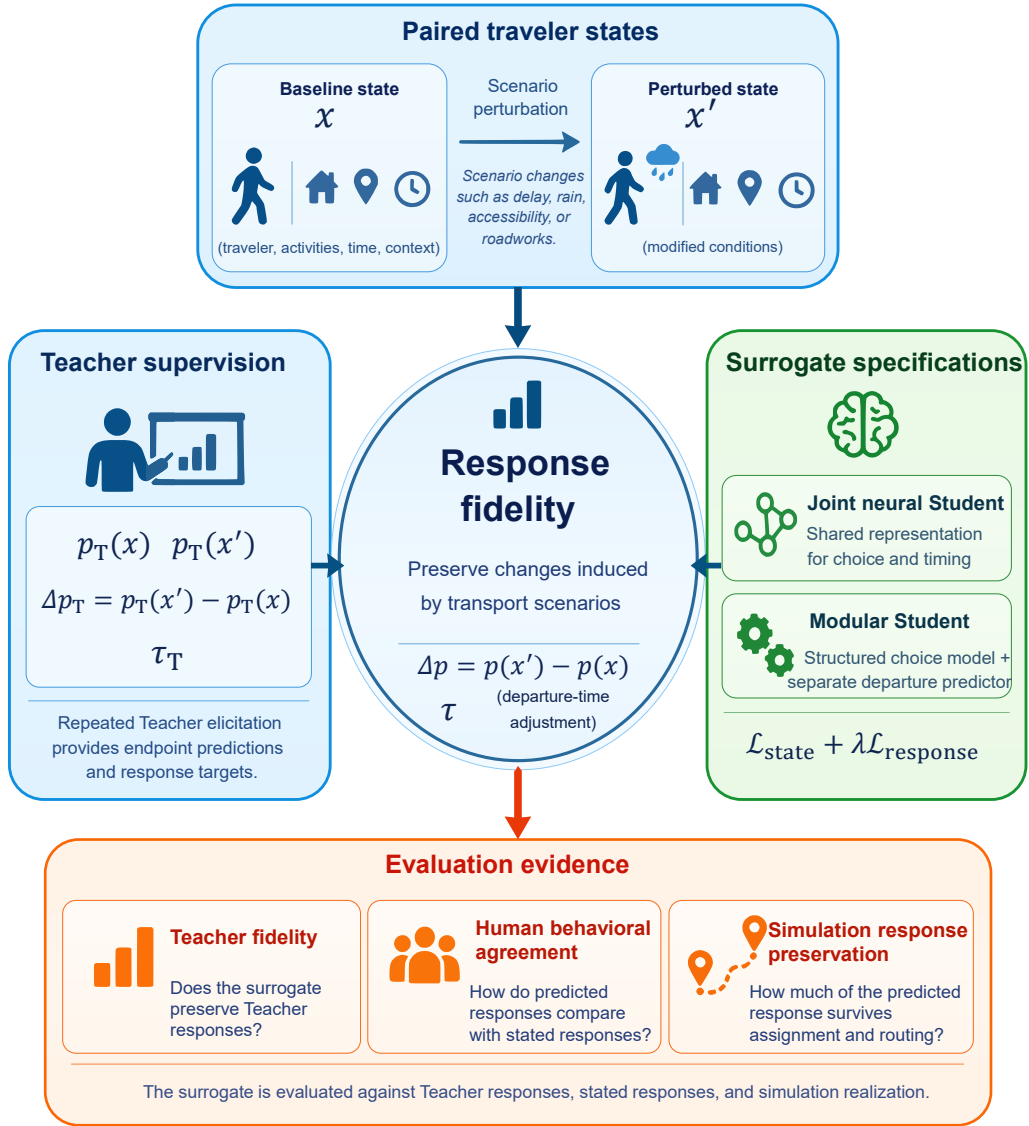


Figure 2: Response-aware behavioral distillation and evaluation. Paired states define changes in mode-choice probabilities, while the Teacher also supplies departure-adjustment targets. Joint neural and modular Students predict choice and timing. The schematic objective combines endpoint fitting with response supervision; the four controlled objectives are specified in Section 3.3, and MNL-S is fitted to endpoint targets. SA-Student additionally uses supply and between-traveler supervision outside that matched comparison. Evaluation distinguishes Teacher fidelity, agreement with stated responses and preservation of predicted responses through assignment, routing and MATSim execution. Arrows link these components conceptually; human responses are used only for evaluation, and simulated experience does not feed back into the behavioral predictions.

3.3. Response-aware learning objectives

Four objectives are compared within the joint architecture. With H_1 denoting the Huber loss with a one-minute transition, the objective at a single state is

$$\ell_{\text{soft}}(\mathbf{x}) = D_{\text{KL}}(\mathbf{p}_T \| \mathbf{p}_S) + H_1(\tau_S - \tau_T). \quad (3)$$

Soft KL applies this loss to both states of every training pair. *CE+KL* adds an equally weighted cross-entropy term for the Teacher's most likely mode, a hard label derived from its probability vector. The two response-aware objectives instead add a penalty on probability changes to soft KL and omit the hard label.

Within a training batch $\mathcal{B}$, let a_{im} indicate that mode m is available in at least one state of pair i . The *signed-response* loss is

$$\ell_{\text{signed}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} |\Delta p_{S,im} - \Delta p_{T,im}|}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im})}. \quad (4)$$

The *direction+magnitude* objective penalizes the sign and the size of the response separately:

$$\ell_{\text{dir}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} b_{im} [-\text{sgn}(\Delta p_{T,im}) \Delta p_{S,im}]_+}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im} b_{im})}, \quad (5)$$

$$\ell_{\text{mag}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} \left| |\Delta p_{T,im}| - |\Delta p_{S,im}| \right|}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im})}, \quad (6)$$

where $b_{im} = \mathbf{1}(|\Delta p_{T,im}| > 10^{-6})$ and $[u]_+ = \max(u, 0)$. Both terms carry unit weight. Their denominators differ because the direction term ignores negligible Teacher responses. The auxiliary supervision of SA-Student plays no part in this comparison.

Once both endpoint targets are given, paired supervision adds no new behavioral information. It changes how much weight the optimization places on scenario contrasts. Writing $\mathbf{e}(\mathbf{x}) = \mathbf{p}_S(\mathbf{x}) - \mathbf{p}_T(\mathbf{x})$ for the endpoint error,

$$\|\Delta \mathbf{p}_S - \Delta \mathbf{p}_T\|_1 \leq \|\mathbf{e}(\mathbf{x}')\|_1 + \|\mathbf{e}(\mathbf{x})\|_1, \quad (7)$$

so a Student that matched the Teacher exactly at every state would also reproduce every response. The experiments ask whether explicit weighting helps when training coverage and model capacity are finite. The penalties themselves do not guarantee agreement in direction. When the two denominators coincide, the direction and magnitude gradients cancel over an interval of oppositely signed responses while endpoint fitting continues, and the hard-label term of CE+KL alters the optimal single-state target (Supplementary Section S2.3). How these idealized properties translate into a shared, finitely trained network is an empirical matter.

3.4. From predicted behavior to simulated demand

Behavioral availability and route feasibility restrict choices in different ways. Availability defines the choice set in Eq. 2, whereas route feasibility depends on whether the route search finds a connection for a given trip and departure time. The PT search considers direct and one-transfer connections, so an unsuccessful search does not prove that no connection exists (Supplementary Section S1).

Three rules convert predicted probabilities into assigned modes. Deterministic assignment selects the most probable mode, stochastic assignment samples from the predicted distribution, and feasibility-constrained assignment samples after removing probability from alternatives for which no route was found and renormalizing the remainder. Without this constraint, a failed route search for car, bicycle or PT triggers a direct fallback to walking, rather than selection of the next-highest-probability mode. The walking route is then constructed on the walking network.

For each traveler, supply attributes are computed at the requested departure time. The fixed model predicts a departure adjustment, which is added before checking route feasibility. The feasibility mask requires behavioral availability and a route returned in the requested mode at this adjusted time. Feasibility-constrained assignment applies this mask and renormalizes once; final routing uses the same adjusted time and routing procedure, without another behavioral prediction. Return trips are routed separately at their scheduled return time, with the same fallback rule. If the retained probability mass is zero, the implementation stops the run rather than dropping that traveler or inventing an alternative. None of the 140 completed runs encountered this condition. Their 140,000 traveler records retain 1,000 people per run, and all recorded outbound and return walking routes contain more than the origin link. Supplementary Section S7 identifies the implementation and the retained decision records.

PT use is then traced through five stages: predicted probabilities, feasibility-adjusted probabilities, assigned modes, routed modes and PT boarding in the simulation. Every stage uses the initial population as its denominator. For a contrast c , the predicted-to-simulated response gap is $G_c = R_c^{\text{sim}} - R_c^{\text{pred}}$. Simulated PT use means boarding a PT vehicle

before reaching the outbound destination, which is not the same as completing the journey. Because the behavioral predictions are held fixed during simulation, the gap reflects assignment, routing and simulation execution with fixed behavioral predictions.

4. Experimental design

Each of the three comparisons in Section 1 draws on a different kind of evidence, summarized in Table 1. The controlled benchmark isolates the effect of the learning objective and compares model specifications against Teacher targets. The stated-choice experiments assess agreement with people on data that play no part in fitting or model selection, and the simulation experiment follows predicted responses through assignment and routing.

Table 1

Evaluation questions, evidence units and comparison scope.

Question	Evidence units	Comparison and scope
RQ1: Teacher-response fidelity	Six held-out synthetic personas; three training seeds	Four neural objectives trained under matched conditions isolate the effect of the loss. MNL-S with separate timing predictors compares model specifications.
RQ2: Stated-response agreement	332 Singapore and 321 Shanghai respondents	Within-person choice contrasts compared with the responses of the fitted models.
RQ2: Teacher comparison	24 respondents per city	Contemporary Teacher and Student responses on the same tasks, decomposed into Teacher and Student discrepancies.
RQ3: Preservation in simulation	1,000 fixed Helsinki travelers	Each model family combined with three assignment rules under a perceived PT delay, simulated in MATSim.

4.1. Controlled Teacher-response benchmark

A persona is a generated traveler profile, and a state combines a persona with a trip, a context, alternative attributes and Teacher targets. The training, validation and test sets contain 1,839, 365 and 437 states drawn from disjoint groups of 28, six and six personas. The test states form 398 response pairs and 94 interaction contrasts. Single-context changes, PT-supply profiles, joint perturbations and context–attribute contrasts all follow the same persona partition (Supplementary Table S7).

Within each of three training seeds, the four neural objectives share the architecture, inputs, availability rules, training pairs, initialization and optimization budget, so that the objective is the only difference between them. Models for the probability-response comparison are selected by macro-source KL, a validation KL divergence summarized at the level of the four supervision sources. MNL-S receives the same endpoint information and exposure weights, is regularized separately and is selected by the same criterion. Optimization settings are reported in Supplementary Section S2.

Departure prediction is evaluated in a separate comparison selected by validation departure MAE. This criterion chooses the bounded-linear and neural timing predictors, the ridge penalty, and independently trained versions of the four joint objectives. The neural and bounded predictors share data, information and update budget, which separates the effect of the timing specification from that of the selection rule.

Response fidelity is measured as the absolute difference between Teacher and Student probability changes, averaged first over the modes available in a pair and then over pairs. Macro-source KL, departure MAE against the Teacher and interaction error complement this measure. Interaction error compares the response to a combined perturbation with the responses to its components. Formal definitions, including the feasibility violation rate, are given in Supplementary Section S2.1.

Two sources of uncertainty are reported. Training-seed standard deviations describe optimization variability, whereas paired intervals clustered by persona describe variation across the six held-out personas and are conditional on them. Consistency within the benchmark is further examined through persona-level results, paired seed differences, removal of each test persona from evaluation without retraining, and reaggregation of the repeated Teacher outputs that are available.

4.2. Stated-choice experiments

The Singapore sample comprises 332 eligible residents who made 3,320 choices. Its ten tasks form five within-person pairs on PT accessibility, rain, fares, PT delay and roadworks. The Shanghai cohort evaluated by the models comprises 321 respondents with 3,208 explicit four-mode choices over ten tasks. Two further answers indicate no suitable mode and are not scored. Twelve eligible respondents whose answers fall outside the model's input mapping are retained in a human-only sensitivity analysis. Responses were collected from 20 August to 5 September 2026 in Singapore and from 12 to 18 September 2026 in Shanghai, and both questionnaires recorded participant consent. We distributed electronic questionnaires through WeChat, WhatsApp and related personal contacts to research-group members' families, friends, classmates and colleagues, and through onward sharing within those networks. Singapore used Google Forms and Shanghai used Wenjuanxing. This convenience and snowball recruitment does not yield representative city samples. Singapore eligibility required residence in Singapore; Shanghai eligibility required travel in Shanghai during the preceding 30 days. Participants completed ten stated-choice tasks online. A privacy statement preceded the survey, and we did not solicit sensitive personal information. The study was conducted on an ethics-exemption basis. Sample flow and questionnaire access are reported in Supplementary Section S1.

The two surveys specify their tasks in different ways. Singapore shows respondents numerical tables of time and cost. Shanghai describes most conditions qualitatively, with explicit values only for a 6-yuan PT fare, a 30-yuan parking fee and one transfer. Model inputs therefore combine attributes displayed in the questionnaire, characteristics reported by respondents and reference values for information that was not collected (Supplementary Table S37). The reference values do not depend on respondents' answers. Twelve alternative specifications for Singapore and eight for Shanghai vary selected missing attributes and reference times (Supplementary Table S4).

The questionnaire's car option is represented by the model's private-car alternative. Private car requires ownership and a licence, and bicycle requires access. All explicit choices are scored, including eight Singapore and two Shanghai choices of a mode that the model treats as unavailable, so that the evaluation sample is not reduced to choices the model can represent.

For respondent i , contrast c , task weights w_{ct} and target mode group M_c , the stated and predicted responses and the response bias are

$$r_{ic}^H = \sum_t w_{ct} \mathbf{1}(y_{it} \in M_c), \quad (8)$$

$$r_{ic}^j = \sum_t w_{ct} \sum_{m \in M_c} p_{itm}^j, \quad j \in \{T, S\}, \quad (9)$$

$$B_c^j = \text{mean}_i(r_{ic}^j - r_{ic}^H). \quad (10)$$

Simple pairs use weights -1 and $+1$, and the rain–delay interaction combines four tasks. Intervals come from a respondent bootstrap that resamples each person's full set of tasks, with shared draws for paired model comparisons. Nominal 95% intervals are reported together with Bonferroni-adjusted intervals over the five Singapore or eight Shanghai contrasts, computed separately for each model and city. These intervals are conditional on the fitted models, and an interval that includes zero does not establish equivalence. Choice accuracy, mode shares, Brier score and negative log-likelihood (NLL) complement the response measures. Because a mode with zero predicted probability makes NLL infinite, NLL is computed with explicit probability truncation (Supplementary Section S3).

A further Shanghai analysis asks whether inputs derived from a transport network improve agreement because they are matched to individual respondents. Origin–destination pairs of approximately 6 km are assigned using OSM roads and a synthetic metro timetable, and the assignment is then repeated and complete supply profiles are exchanged between respondents. The constructed trips test sensitivity to numerical inputs and are not observations of respondents' journeys.

4.3. Contemporary Teacher comparison

To compare Teacher–human and Student–Teacher discrepancies on the same tasks, the Teacher is queried directly on the stated-choice tasks for 24 respondents per city, stratified by age and car ownership. Each of the ten tasks per respondent receives three successful Teacher elicitations. The Teacher receives the same numerical and categorical attributes and the same availability rule as the Students, but not the human answers. Metadata that the Students do not use are set to neutral values, and the prompt explains PT feasibility and its time components. Service settings and repeat variability are reported in Supplementary Section S3.2.

On the same respondents, the signed response discrepancy of a Student decomposes as

$$\bar{r}_c^S - \bar{r}_c^H = (\bar{r}_c^T - \bar{r}_c^H) + (\bar{r}_c^S - \bar{r}_c^T). \quad (11)$$

The first term is the Teacher's own departure from stated behavior, and the second is the difference between Student and Teacher. The decomposition is descriptive. Because the Teacher is queried after the Students were trained, the second term may also contain changes in the remotely served model over that period, a form of drift documented for LLM survey simulation more generally (Bisbee et al., 2024). All terms use common respondent resamples, and Teacher repeat variability is reported separately.

4.4. Simulation and computational experiments

The Helsinki experiment compares a baseline and a perceived-delay condition for a fixed population of 1,000 travelers, using OSM roads and HSL timetables. The delay condition adds 15 minutes to the PT delay perceived by the behavioral model while leaving the physical timetable unchanged. SA-Student, the soft-KL and direction+magnitude neural Students, and MNL-S with independent neural timing are each combined with the three assignment rules of Section 3.4. Common random numbers are used across paired conditions and assignment rules, and training and assignment variability are summarized separately (Supplementary Table S26).

At each stage, the response R is the change in PT probability or PT-use share from baseline to delay over the same 1,000 people. A single MATSim iteration with fixed behavioral predictions measures the gap G introduced by assignment and routing. The design therefore does not represent adaptation to an actual service disruption or a day-to-day equilibrium. Additional comparisons route trips at desired and at adjusted departure times. Walking and cycling use network-derived link speeds, and a Shanghai experiment examines alternative active-mode speed assumptions while keeping people, assigned modes, departure times and plans fixed.

A Singapore experiment records the time needed to construct simulation scenarios for nested populations of up to 50,000 agents, separating behavioral inference from feature and route preparation and measuring construction with and without reuse of unchanged supply calculations. Teacher latency is measured on successful requests only, and population-scale Teacher times are projections rather than measured builds (Supplementary Sections S5 and S6).

5. Results

5.1. Response fidelity and behavioral specification

Explicit response supervision improves the property it targets. Direction+magnitude lowers mean response error from 0.06930 under soft KL to 0.06586, a reduction of 4.97% (Table 2; Figure 3a). The paired difference is -0.00345 , with a persona-clustered 95% interval of $[-0.00504, -0.00191]$. Signed response reaches 0.06653. The difference between the two paired objectives is small (-0.00067 , interval $[-0.00199, 0.00072]$), which supports explicit response supervision but does not establish an additional benefit from separating sign and size.

The improvement is small but consistent. Across the three training seeds, direction+magnitude differs from soft KL by -0.00499 , -0.00441 and -0.00094 , although the interval for the third seed includes zero. Averaged over seeds, the error is lower for each of the six test personas, and removing any one persona from evaluation leaves differences between -0.00408 and -0.00310 (Supplementary Tables S11 and S13). The advantage of both paired objectives also persists when the targets are recomputed from the available repeated Teacher outputs, an analysis that excludes the mechanism states for which repeats were not recovered.

The benefit is specific to single contrasts. For combined perturbations, the paired objectives reach interaction errors of 0.0465 and 0.0464, slightly above the 0.0457 of soft KL, so supervising individual responses did not improve interaction error in this benchmark. CE+KL has the largest response error, 0.07528, even though it fits an additional target. Its hard label pulls predictions toward the Teacher's leading mode, and in the fitted networks this enlarges the probability changes on pairs whose leading mode does not change (Supplementary Section S2.3). Fitting the most likely mode appears to distort the graded responses on which scenario analysis depends.

A change of specification matters more than any change of objective. MNL-S, fitted to the same endpoints with linear utilities, attains the lowest errors on all three measures: 0.05644 for response error, 0.07310 for macro-source KL and 0.03865 for interaction error. Its response error is 14.30% below that of direction+magnitude, with a paired interval of $[-0.01246, -0.00722]$. One plausible reading is that structured utilities restrict how a change in an attribute can propagate to probabilities. With only 28 training personas, this restriction appears to generalize better than the

Table 2

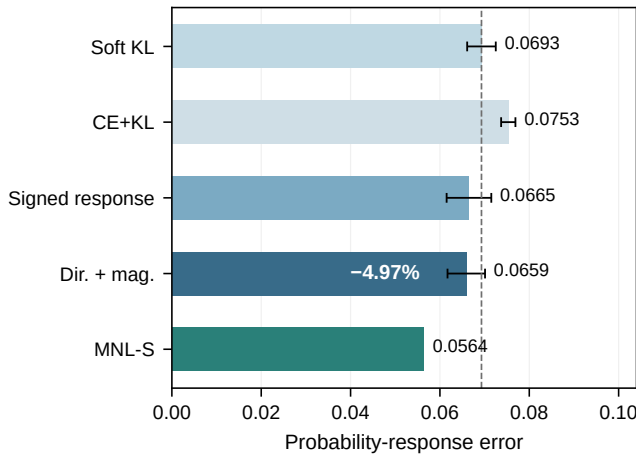
Probability-response fidelity on the held-out Teacher benchmark.

Model / objective	Macro-source KL ↓	Response error ↓	Interaction error ↓
Soft KL	0.0915 ± 0.0187	0.0693 ± 0.0032	0.0457 ± 0.0038
CE+KL	0.1046 ± 0.0170	0.0753 ± 0.0016	0.0518 ± 0.0038
Signed response	0.0871 ± 0.0138	0.0665 ± 0.0050	0.0465 ± 0.0036
Direction+magnitude	0.0861 ± 0.0145	0.0659 ± 0.0042	0.0464 ± 0.0039
MNL-S	0.0731	0.0564	0.0386

Neural entries are means $\pm$ SD over three training seeds. The test set contains 437 states, 398 response pairs and 94 interaction contrasts from six held-out synthetic personas. All models, including the MNL-S regularization, are selected by validation macro-source KL. MNL-S uses the same inputs and training states as the neural Students.

(a) Response supervision

Models selected by validation KL

**(b) Modular departure prediction**

Fixed MNL-S choice probabilities

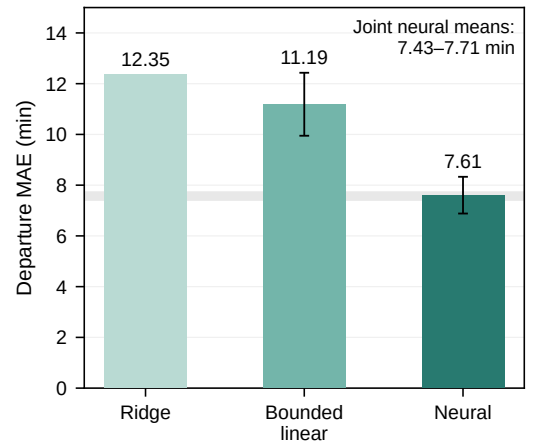


Figure 3: Response learning and modular departure prediction. (a) Probability-response errors for models selected by validation KL. The dashed line marks soft KL. (b) Departure MAE of three timing predictors combined with the same MNL-S choice model, whose probabilities are identical across the three. The shaded band spans the means of the four joint neural models selected for timing (7.43–7.71 min). Error bars show training-seed SD where applicable. Departure targets are the Teacher’s.

Table 3

Departure prediction under a common selection criterion (validation MAE).

Choice model	Departure predictor	Total parameters	MAE (min) ↓
Joint / soft KL	Shared neural encoders	24,562	7.61 ± 0.75
Joint / CE+KL	Shared neural encoders	24,562	7.43 ± 0.91
Joint / signed response	Shared neural encoders	24,562	7.71 ± 0.66
Joint / direction+magnitude	Shared neural encoders	24,562	7.55 ± 0.80
MNL-S	Ridge regression	555	12.35
MNL-S	Bounded linear	555	11.19 ± 1.24
MNL-S	Independent neural	18,733	7.61 ± 0.72

Errors are measured against Teacher departure targets on the 437 test states. Neural and bounded-linear predictors report means $\pm$ SD over three training seeds, and the ridge fit is deterministic. MNL-S choice probabilities are identical across the three timing predictors. The joint models in this table are trained separately and selected by validation departure MAE, whereas those in Table 2 are selected by validation KL.

additional freedom of the neural Students, which can fit individual states without reproducing the contrasts between them.

The linear specification is less suited to timing. With the MNL-S choice probabilities held fixed, ridge regression, the bounded linear predictor and the independent neural predictor give departure MAEs of 12.35, 11.19 and 7.61 minutes (Table 3; Figure 3b). The neural predictor brings the modular model within the 7.43–7.71-minute range of the joint models selected for timing, with 18,733 rather than 24,562 parameters. Choice and timing therefore favor different functional forms, and estimating them separately retains the response advantage of linear utilities with comparable departure errors in this benchmark. All timing errors are measured against Teacher targets, not observed departure behavior.

5.2. Agreement with stated responses

How well a Student agrees with people depends on the intervention. In Singapore, SA-Student predicts a PT-delay response of -14.86 percentage points, compared with a stated change of -12.05 points, a bias of -2.81 points (nominal 95% interval $[-8.18, 2.75]$). Rain lowers its bicycle-plus-walk probability by 19.70 points, close to the 21.99-point fall in stated choice. Poorer PT accessibility is a different case. The Student predicts a change of -14.42 points against a stated -24.40 , and the resulting $+9.98$ -point bias remains positive after adjustment for multiple contrasts ($[3.01, 16.93]$).

Shanghai reverses the pattern for delay. Under the reference specification, SA-Student predicts a PT-delay response of -30.57 points, roughly twice the stated -14.64 . The bias of -15.93 points excludes zero after adjustment over eight contrasts ($[-22.41, -9.21]$). The Student also predicts responses of the wrong sign for the fare and walking-versus-waiting contrasts (Figure 4). These discrepancies persist across the eight alternative input specifications, where the delay bias ranges from -16.53 to -10.25 points and both sign reversals remain. These specifications retain the 15-minute encoding of the qualitative delay condition; they do not test sensitivity to its intensity.

The ranking of objectives on the Teacher benchmark does not carry over to people. In Singapore, direction+magnitude shifts the delay bias from $+1.78$ points under soft KL to -1.34 , yet both objectives substantially understate the rain response. MNL-S offers the clearest example. It approximates the Singapore fare response and the Shanghai parking and road responses, but under poorer Singapore accessibility it predicts a PT change of $+13.85$ points, whereas the stated PT share falls by 24.40 points. The accessibility task raises PT travel time from 32 to 48 minutes, access walking from three to twelve minutes and transfers from zero to two (Supplementary Table S2). The reversal thus concerns this combined access burden rather than any single walking-time coefficient. The model with the smallest Teacher error is, for this intervention, the only one that predicts the wrong direction. Conversely, CE+KL, which has the largest Teacher-response error, comes closest to the stated accessibility response (-23.37 against -24.40 points; Supplementary Table S16).

Response bias and level bias also rank the models differently (Table 4). SA-Student underpredicts the PT share by 30.67 points in Singapore and 16.84 in Shanghai, whereas MNL-S narrows these gaps to 3.77 and 5.36 points. In Singapore, however, MNL-S has the larger mean absolute response bias, 11.1 points against 4.0 for SA-Student. A model can therefore reproduce how many people use PT while misrepresenting how that number changes, and the reverse also occurs. Accuracy is a weak guide in either case. SA-Student's accuracy of 66.93% in Singapore barely exceeds the 66.57% obtained by always predicting PT, while in Shanghai it reaches 80.46%.

Numerical inputs affect agreement as well, though not mainly through respondent-specific matching. In Shanghai, replacing reference values with network-derived inputs raises accuracy by 2.80 percentage points, and the gain remains 2.56 points after assigning fresh origin–destination pairs and 2.52 points after exchanging complete supply profiles between respondents. Most of the improvement therefore comes from a general shift in the numerical inputs rather than from matching them to individual respondents (Supplementary Section S3.4).

5.3. Teacher–human and Student–Teacher discrepancies

The matched 24-person subsets describe two components of discrepancy under the reference inputs (Figure 5). They compare current predictions and do not identify the historical origin of the differences. The first component is the contemporary Teacher–human discrepancy. Under poorer accessibility, the selected Singapore respondents reduce PT choice by 37.50 points, while the contemporary Teacher predicts a reduction of 15.40 points and SA-Student 13.46. The attenuation is also observed in the contemporary Teacher, and SA-Student differs from it by $+1.94$ points to it (nominal 95% interval $[-3.40, 7.26]$).

The second component is the Student–Teacher discrepancy. For the encoded Shanghai delay task, the selected respondents and the Teacher agree closely, at -8.33 and -8.17 points, but SA-Student predicts -34.87 . Its difference

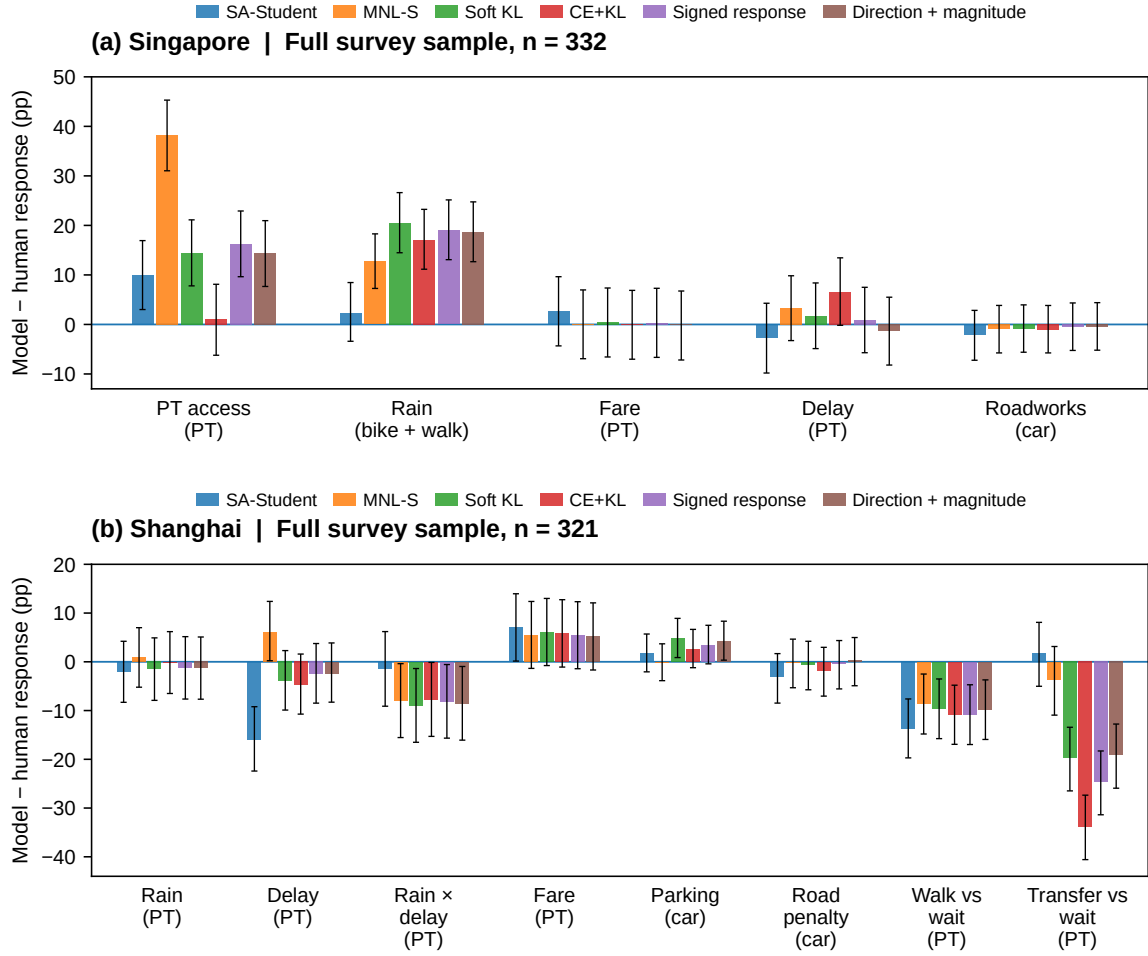


Figure 4: Response differences in the full samples, defined as the change in model probability minus the change in stated-choice share. Whiskers show Bonferroni-adjusted respondent-bootstrap intervals within each city (99% for Singapore and 99.375% for Shanghai). Neural means average three fitted seeds. Shanghai results use the reference specification of Section 4.2. The interaction and walking-versus-waiting contrasts include 320 respondents.

Table 4

Choice accuracy, PT share bias and mean absolute response bias against stated choices.

Model	Singapore ($n = 332$)			Shanghai ($n = 321$)		
	Acc. (%)	PT bias (pp)	Mean $ B_c $	Acc. (%)	PT bias (pp)	Mean $ B_c $
SA-Student	66.9	-30.7	4.0	80.5	-16.8	5.8
MNL-S	71.1	-3.8	11.1	81.8	-5.4	4.1
Soft KL	72.9 ± 0.9	-17.7 ± 3.3	8.2 ± 3.3	78.5 ± 1.7	-12.3 ± 7.9	7.6 ± 0.9
CE+KL	69.7 ± 0.3	-23.5 ± 3.5	6.9 ± 1.7	77.1 ± 3.0	-15.0 ± 6.8	9.1 ± 1.9
Signed response	73.1 ± 1.2	-16.0 ± 3.5	8.6 ± 3.0	79.8 ± 1.1	-10.7 ± 6.2	7.7 ± 1.8
Direction+magnitude	71.8 ± 3.1	-16.7 ± 6.2	8.8 ± 2.0	80.3 ± 0.5	-10.4 ± 5.9	7.5 ± 1.0

All 3,320 Singapore and 3,208 Shanghai explicit choices are scored, including eight and two choices of modes the model treats as unavailable. PT bias is the mean predicted PT probability minus the stated PT share. Mean $|B_c|$ is computed within each fitted seed by taking the absolute bias for each contrast and averaging over the five Singapore or eight Shanghai contrasts; these seed-level metrics are then averaged. Figure 4 instead plots signed biases averaged over seeds, whose absolute values need not equal the table metric. Both are in percentage points. Neural entries are means $\pm$ SD over three training seeds. Always predicting PT gives accuracies of 66.57% and 71.98%. Further metrics are given in Supplementary Table S15.

from the Teacher is -26.70 points, with a family-adjusted interval of $[-36.95, -15.42]$. The other Students respond less strongly, at -19.03 for soft KL, -17.17 for direction+magnitude and -8.85 for MNL-S, so SA-Student shows the largest discrepancy in this comparison. This pattern does not isolate its training history from changes in the Teacher service.

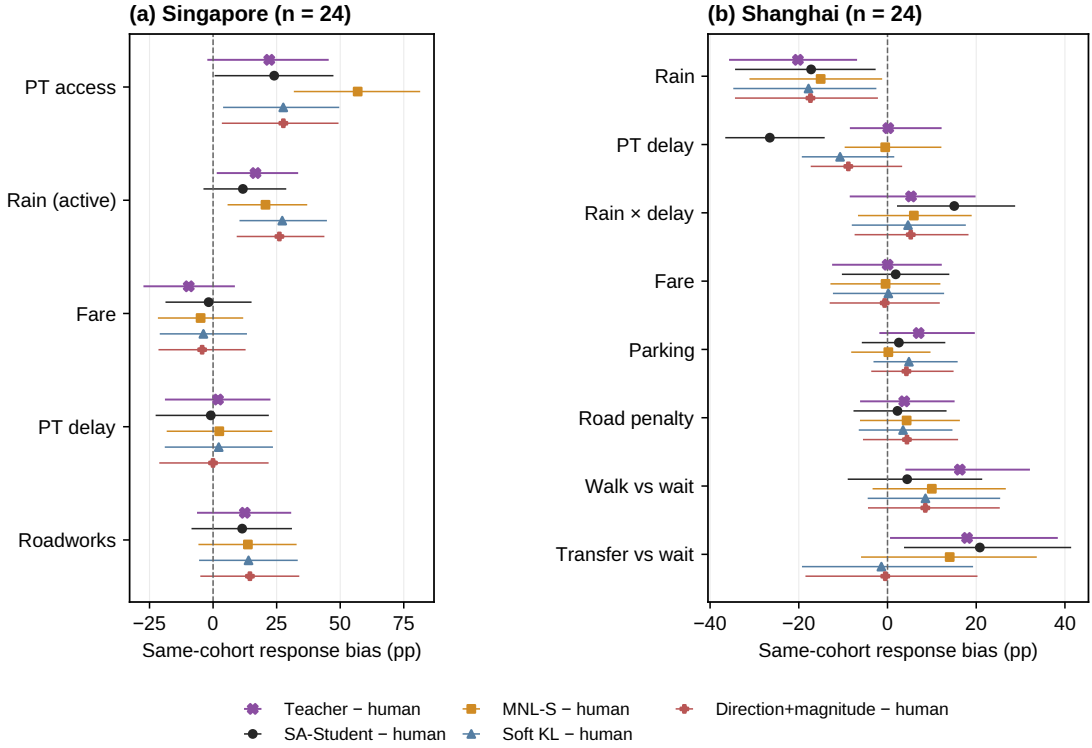


Figure 5: Teacher and Student response differences on the matched 24-person subsets. Points show the change in model probability minus the change in stated-choice share, and bars show nominal 95% respondent-bootstrap intervals. Teacher means average three elicitations per state, and controlled neural means average three fitted seeds. Paired intervals between Student and Teacher are given in Table 5 and the supplement.

The two terms of Eq. 11 can also offset each other (Table 5). In the Shanghai walking-versus-waiting contrast, the Teacher exceeds the human response by $+16.31$ points and SA-Student falls 11.88 points below the Teacher, leaving a total bias of only $+4.43$ points. A small total bias can therefore conceal two larger, opposing discrepancies. For this reason the decomposition is more informative than the total bias for distinguishing the two observed discrepancies, although the Student–Teacher term may also include changes in the Teacher since training.

Table 5

Decomposition of signed response differences among stated choices, the contemporary Teacher and SA-Student.

Contrast	Teacher – human	SA-Student – Teacher	SA-Student – human
<i>Singapore (n = 24)</i>			
Poorer PT access	+22.1 [−9.9, 52.1]	+1.9 [−4.9, 9.1]	+24.0 [−7.2, 54.3]
Rain (active modes)	+16.7 [−2.4, 38.4]	−5.0 [−11.4, 1.1]	+11.7 [−8.0, 33.9]
PT fare	−9.6 [−33.4, 14.8]	+7.8 [4.1, 12.4]	−1.8 [−26.2, 22.6]
PT delay	+1.9 [−25.2, 28.9]	−2.9 [−12.3, 5.8]	−0.9 [−28.8, 29.7]
Road penalty	+12.5 [−12.0, 36.3]	−1.0 [−5.8, 4.2]	+11.4 [−14.8, 37.0]
<i>Shanghai (n = 24)</i>			
Rain	−20.2 [−41.6, −2.5]	+3.0 [−2.9, 8.0]	−17.2 [−41.5, 2.3]
PT delay	+0.2 [−10.2, 18.0]	−26.7 [−36.9, −15.4]	−26.5 [−39.3, −8.3]
Rain × delay	+5.3 [−13.9, 26.1]	+9.7 [3.3, 16.2]	+15.0 [−2.5, 34.7]
PT fare	+0.0 [−16.8, 16.7]	+1.8 [0.3, 3.2]	+1.8 [−14.6, 18.3]
Parking	+7.0 [−3.0, 24.6]	−4.5 [−11.0, −0.3]	+2.6 [−8.1, 17.7]
Road penalty	+3.8 [−10.3, 20.1]	−1.5 [−4.7, 0.7]	+2.2 [−11.6, 18.3]
Walk versus wait	+16.3 [0.2, 39.6]	−11.9 [−14.2, −9.1]	+4.4 [−13.0, 29.2]
Transfer versus wait	+17.9 [−7.4, 44.4]	+2.9 [0.4, 5.2]	+20.8 [−4.0, 46.8]

Entries are percentage points with Bonferroni-adjusted respondent-bootstrap intervals, computed within each city over five Singapore or eight Shanghai contrasts. All terms use the same respondents, the same fitted SA-Student and Teacher means over three elicitations. Point estimates are rounded to one decimal place. The SA-Student – Teacher term may include changes in the Teacher since training.

5.4. Response preservation in simulation

For SA-Student in Helsinki, the predicted response to perceived PT delay is −13.10 percentage points. Deterministic assignment gives −13.00 points, and routed modes and simulated PT use both give −12.30. The resulting gap of +0.80 points has a paired-person 95% interval of [−0.77, 2.38]. The level of PT use behaves very differently. At baseline, mean PT probability is 27.74%, 218 of the 1,000 travelers are assigned PT, and 161 board PT in the simulation, so 73.85% of assigned PT users board before reaching the outbound destination. PT use thus falls by more than eleven points between prediction and simulation, while the delay response changes by less than one point. The distinction between levels and responses that separates static from response fidelity thus reappears at the network stage.

Stochastic assignment represents the predicted distribution in expectation, but a finite population still varies from draw to draw. For SA-Student, ordinary and feasibility-constrained sampling both give a mean simulated response of −14.17 points, with assignment SDs of 1.55 and 1.40 points. Feasibility adjustment changes the probability response to −12.83 points and prevents PT assignments for which the specified search finds no route. It makes the assigned demand routable, but it does not bring the simulated response closer to the original prediction in this population. Routability and response preservation are separate properties of an assignment rule.

MNL-S with neural timing predicts −6.84 points and yields −6.90 under deterministic assignment (Table 6). Its stochastic and feasibility-constrained means are −6.63 and −6.67. For a given assignment seed, PT use is identical across its three independently fitted timing predictors, so timing variability does not alter this model's simulated mode response.

Signed averages can hide opposing errors. The stochastic-assignment gap of soft KL averages approximately zero across fitted models, but the model-specific means range from −2.17 to +2.18 points. When assignment repetitions are averaged within each fitted model before taking absolute values, the mean absolute gap is 1.45 points for soft KL and 0.91 for direction+magnitude (Supplementary Table S28). In this summary, the objective that preserves Teacher responses better also reaches the network with smaller model-specific gaps, a difference that the signed mean conceals.

Departure timing affects routing even when feasibility does not change. Applying SA-Student's predicted departure adjustment alters the selected itinerary in 279 baseline and 313 delay cases while leaving binary PT feasibility unchanged. The timing output therefore shapes the schedule that travelers meet in the simulation, an effect that a feasibility check alone would not reveal.

Journey times depend on the representation of the network as well as on mode counts. In the Shanghai speed experiment, PT counts stay at 224 at baseline and 134 under delay, and all 321 people complete their journeys. The

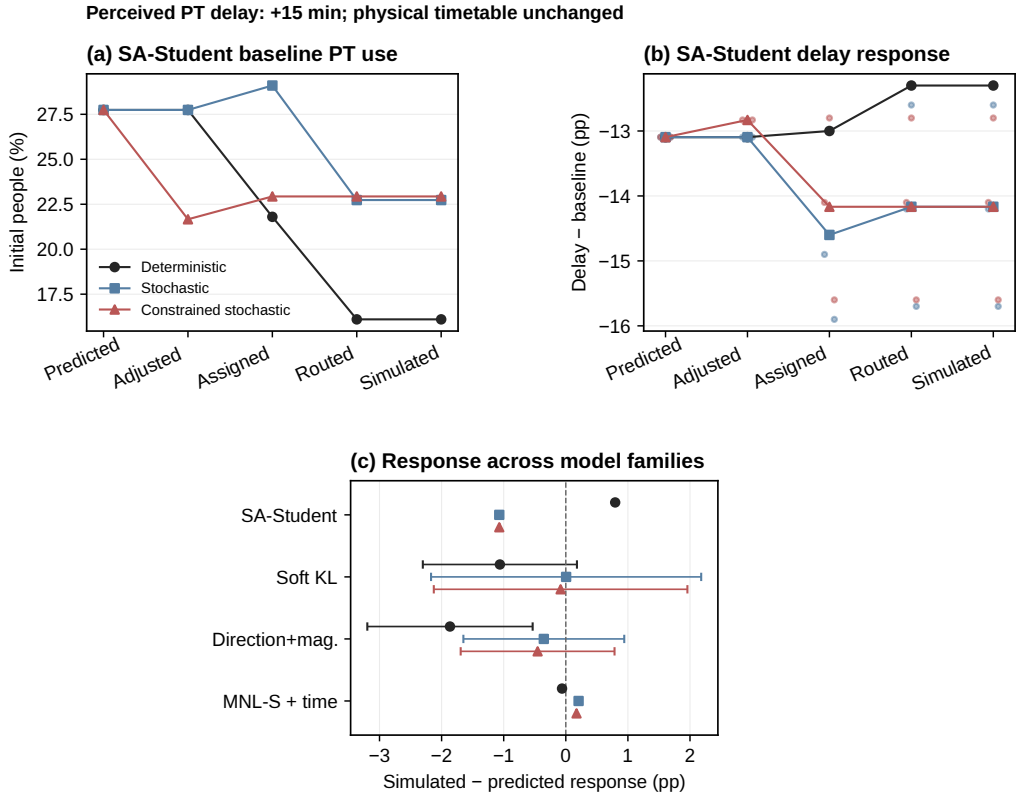


Figure 6: Response preservation for a fixed population of 1,000 Helsinki travelers under a 15-minute perceived PT delay, with the physical timetable unchanged. (a,b) Baseline PT shares and delay responses of SA-Student across predicted probabilities, feasibility-adjusted probabilities (Adjusted), assigned modes, routed modes and simulated PT use. Small points in (b) show individual assignment seeds. (c) Predicted-to-simulated gaps by model family. Error bars show training-seed SD after averaging assignment repetitions within each model. Simulated PT use is defined by boarding and does not require journey completion.

paired change in outbound time is nevertheless -4.44 minutes with network-derived active-mode speeds and $+14.04$ minutes with mode-specific speed caps for walking and cycling, the latter with an interval of $[11.18, 17.05]$ minutes. Identical demand plans can thus imply time changes of opposite sign (Supplementary Section S5.2).

Table 6

Predicted and simulated PT responses in the fixed 1,000-person Helsinki population.

Model	Assignment	Predicted (pp)	Simulated (pp)	Gap (pp)	Assignment SD
SA-Student	Deterministic	-13.10	-12.30	0.80	–
SA-Student	Stochastic	-13.10	-14.17	-1.07	1.55
SA-Student	Constrained	-13.10	-14.17	-1.07	1.40
Soft KL	Deterministic	-6.97 ± 4.92	-8.03 ± 5.13	-1.06 ± 1.24	–
Soft KL	Stochastic	-6.97 ± 4.92	-6.97 ± 3.32	0.01 ± 2.18	0.27
Soft KL	Constrained	-6.97 ± 4.92	-7.06 ± 3.41	-0.08 ± 2.04	0.28
Direction+magnitude	Deterministic	-6.77 ± 4.96	-8.63 ± 4.76	-1.87 ± 1.33	–
Direction+magnitude	Stochastic	-6.77 ± 4.96	-7.12 ± 3.97	-0.35 ± 1.30	0.35
Direction+magnitude	Constrained	-6.77 ± 4.96	-7.22 ± 4.06	-0.45 ± 1.24	0.32
MNL-S + neural time	Deterministic	-6.84 ± 0.00	-6.90 ± 0.00	-0.06 ± 0.00	–
MNL-S + neural time	Stochastic	-6.84 ± 0.00	-6.63 ± 0.00	0.21 ± 0.00	1.10
MNL-S + neural time	Constrained	-6.84 ± 0.00	-6.67 ± 0.00	0.17 ± 0.00	1.21

Responses are changes in PT use from baseline to perceived delay over the same 1,000 people, with the physical timetable unchanged. Gap is the simulated minus the predicted response. Constrained denotes feasibility-constrained stochastic assignment. Values average assignment repetitions within each fitted model and then across models, and $\pm$ gives the SD across three training seeds. SA-Student is a single fitted model. Assignment SD is the mean within-model SD across assignment repetitions. Intermediate stages and absolute-gap summaries are given in Supplementary Tables [S27–S28](#).

5.5. Computational implications for scenario analysis

Once the Teacher is distilled, behavioral prediction ceases to be the main computational cost. Constructing a scenario for 10,000 Singapore agents takes 51.1 minutes, of which behavioral prediction accounts for 10.3 seconds. For 50,000 agents the corresponding times are 231.8 minutes and 48.8 seconds, so inference contributes approximately 0.34% and 0.35% of construction time. In a model-only benchmark that excludes feature encoding and routing, MNL-S with neural timing requires 1.68 ± 0.33 microseconds per state, compared with 1.78–2.16 microseconds for the joint neural objectives.

The dominant cost lies in network and timetable calculations. Reusing those that do not change between scenarios reduces construction time from 3,223.6 to 469.5 seconds in Singapore and from 10,491.8 to 103.2 seconds in Helsinki, with mode and origin–destination assignments unchanged. Each city was measured under one reuse condition, and concurrent workload limits the precision of these timings. The differences among compact Students are negligible by comparison.

These measurements concern repeated prediction after distillation. Teacher latency was measured on successful requests, and population-scale Teacher times are projections. A full cost comparison would also need to account for the original supervision, failed requests, retries and training (Supplementary Section [S6](#)).

6. Discussion

6.1. What a distilled traveler model should preserve

The benchmark results support treating responses, rather than predictions at individual states, as the object of behavioral distillation. Relative to soft KL, direction+magnitude supervision lowered Teacher-response error in every seed and, on average over seeds, for every held-out persona. The gain was modest, however, and did not extend to combined perturbations. The inequality in Eq. 7 relates endpoint errors to response errors; it does not determine the size of the observed gain. Paired supervision does not add information beyond the endpoint targets, it only redirects optimization toward the contrasts that a scenario queries. Its value can be expected to matter most where training coverage is thin relative to the variety of scenarios, which is the usual situation when an LLM supplies the supervision.

The larger lesson concerns specification. A linear utility model distilled from the same Teacher targets preserved responses better than any neural objective, while a separately trained neural predictor was needed for departure timing. Choice and timing thus appear to call for different functional forms, and a modular Student lets each be estimated and later recalibrated on its own terms. A linear choice model also keeps the distilled behavior open to inspection, since its

utilities can be compared with established evidence on time and cost sensitivity. This property is relevant to the hybrid LLM and conventional modeling that has been proposed for transport agent-based simulation (Liu et al., 2025), where the LLM would supply behavioral knowledge and a transparent model would carry it into the simulation.

6.2. Teacher fidelity and agreement with people

Fidelity to the Teacher did not predict agreement with stated behavior. MNL-S, the most faithful Student, reversed the stated response to poorer accessibility in Singapore, whereas CE+KL, the least faithful objective, came closest to it. A related dissociation appeared between levels and responses. SA-Student understated PT use by about 31 points in Singapore yet had the smallest average response bias there, while MNL-S came within four points of the PT share yet reversed the accessibility response and understated the rain response. These patterns are consistent with evidence that LLMs capture the direction of travel trade-offs more reliably than their magnitude (Yan et al., 2026), and they extend that concern from the LLM to the models distilled from it.

The contemporary Teacher comparison separates discrepancies that call for different follow-up analyses. For Singapore accessibility, closer imitation alone cannot eliminate the remaining Teacher–human discrepancy. Matching the current Teacher would reduce SA-Student’s absolute response bias from 24.04 to 22.10 percentage points on this subset, leaving a substantial difference from stated behavior. Prompt design, input representation and calibration against observed choices are possible avenues to examine. For the Shanghai delay task under the reference encoding, the current Teacher is close to the human point estimate while SA-Student departs more strongly. This motivates checking the Student’s specification and training history, but the comparison cannot exclude changes in the Teacher service. Cancellation between the two terms, as in the Shanghai walking-versus-waiting contrast, shows that a small total bias is not in itself evidence of a sound model. Calibration to stated behavior is a task distinct from compression, and the surveys analyzed here were used only for evaluation.

6.3. From predicted responses to simulated demand

Assignment and routing reshape predicted behavior before it becomes demand. In Helsinki, PT use fell by more than eleven points between prediction and simulation for SA-Student, while its delay response moved by less than one point. A simulation can therefore preserve a response well and still misstate the level of demand on which that response acts, and the two quantities deserve separate reporting. Feasibility-constrained assignment illustrates a related point. It removed unroutable PT assignments without narrowing the response gap, so improving the consistency of assigned demand with the network does not by itself improve preservation of the behavioral prediction.

Reporting choices matter as well. Signed average gaps near zero concealed model-specific gaps of about two points in either direction, and absolute summaries revealed a difference between objectives that the signed mean hid. Departure adjustments changed selected itineraries without changing feasibility, and identical demand plans in Shanghai produced journey-time changes of opposite sign under different active-mode speed assumptions. Retaining probabilities and departure adjustments until assignment, and reporting each stage against the same population, makes these transformations visible and separable from the behavioral model.

6.4. Scope and further evaluation

The benchmark rests on one Teacher and six held-out synthetic personas. Additional training seeds would describe optimization variability more precisely but would not widen persona coverage, and a second Teacher would be needed to show which findings are specific to the LLM used here. The stated-choice samples were collected over short periods through convenience and snowball recruitment within personal networks, and the two cities used different task formats. Differences between Singapore and Shanghai therefore cannot be attributed to population preferences alone, and the contemporary Teacher comparison rests on 24 respondents per city.

Model inputs for the surveys combine displayed attributes, reported characteristics and researcher-specified values. The sensitivity analyses cover selected defaults and reference times while holding the Shanghai delay encoding at 15 minutes, and monetary values were supplied to the Teacher without explicit currency names. Agreement with stated choices should accordingly be read as agreement under these specifications rather than as calibrated forecasting accuracy. The Helsinki experiment varies perceived delay with the timetable held fixed and uses a single MATSim iteration without behavioral re-estimation. Physical service disruptions, feedback from experienced congestion and equilibrium forecasting remain to be examined. The retained research records extend beyond the manuscript package, while the linked repository currently requires permission to access. Supplementary Section S7 distinguishes retained records, repository contents and unresolved historical request metadata.

7. Conclusion

LLM-derived traveler models can be judged against their Teacher, against people and against the demand that reaches a simulated network. This study shows that these three comparisons need not agree. Supervising paired probability changes improves fidelity to the Teacher's responses, and a linear utility Student combined with a separate neural departure predictor improves it further while keeping departure accuracy close to that of joint neural models. The same models, however, rank differently when compared with stated responses in Singapore and Shanghai, and the network stage in Helsinki transforms the level of demand far more than the predicted response.

For practice, distillation makes LLM-derived behavior cheap enough that behavioral prediction is no longer the computational bottleneck of scenario construction. The more demanding task is evaluation. A traveler model intended to inform a particular intervention should be tested on that intervention, against both its Teacher and human evidence, and its predicted response should be traced through assignment and routing before the simulated outcome is interpreted. Extending this evaluation to further Teachers, broader populations and physical supply disruptions is the natural next step.

Data and code availability

The manuscript package provides model and input definitions, the accessibility Teacher prompt, SA-Student adaptation settings, editable figures and summary data. The research repository is https://github.com/AnXu-ITS/DeepSeek_Traveler_Behavior_Distillation_Research. It currently requires access permission and contains source code, released model checkpoints and selected supporting data. The authors also retain the training targets, evaluation predictions, statistical scripts and detailed records for the later experiments, including the 140 Helsinki runs. These later records are not all present in the linked repository. Requests to inspect supporting materials should be directed to the corresponding author, subject to participant privacy and third-party data terms. Supplementary Section S7 distinguishes the manuscript package, repository contents and retained local records.

References

- Aher, G.V., Arriaga, R.I., Kalai, A.T., 2023. Using large language models to simulate multiple humans and replicate human subject studies, in: Proceedings of the 40th International Conference on Machine Learning, pp. 337–371.
- Argyle, L.P., Busby, E.C., Fulda, N., Gubler, J.R., Rytting, C., Wingate, D., 2023. Out of one, many: Using language models to simulate human samples. *Political Analysis* 31, 337–351.
- Axhausen, K.W., 2002. Some ideas for agent-based transport simulation. *Arbeitsberichte Verkehrs-und Raumplanung* 125.
- Ben-Akiva, M.E., Lerman, S.R., 1985. Discrete choice analysis: theory and application to travel demand. volume 9. MIT press.
- Bisbee, J., Clinton, J.D., Dorff, C., Kenkel, B., Larson, J.M., 2024. Synthetic replacements for human survey data? The perils of large language models. *Political Analysis* 32, 348–362.
- Chen, T., Shen, Z., Zhou, B., Liu, Y., Wang, S., Ke, J., 2026. Addressing the online incremental transport mode choice prediction problem with an LLM-augmented class-incremental learning approach. *Transportation Research Part C: Emerging Technologies* 188, 105709. doi:10.1016/j.trc.2026.105709.
- Goli, A., Singh, A., 2024. Frontiers: Can large language models capture human preferences? *Marketing Science* 43, 709–722.
- Gou, J., Yu, B., Maybank, S.J., Tao, D., 2021. Knowledge distillation: A survey. *International Journal of Computer Vision* 129, 1789–1819.
- Hensher, D.A., Greene, W.H., 2003. The mixed logit model: the state of practice. *Transportation* 30, 133–176.
- Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.
- Horni, A., Nagel, K., Axhausen, K.W., 2016. The multi-agent transport simulation MATSim. Ubiquity Press London.
- Liu, Q., Li, C., Ma, W., 2026a. GATSim: Urban mobility simulation with generative agents. *Transportation Research Part C: Emerging Technologies* 186, 105576. doi:10.1016/j.trc.2026.105576.
- Liu, Q., Li, C., Ma, W., 2026b. Generative agents for urban mobility: A cognitive framework for realistic travel behavior simulation. *Simulation Modelling Practice and Theory* 147, 103234. doi:10.1016/j.simpat.2025.103234.
- Liu, T., Yang, J., Yin, Y., 2025. Toward llm-agent-based modeling of transportation systems: A conceptual framework. *Artificial Intelligence for Transportation* 1, 100001. doi:10.1016/j.ait.2025.100001.
- Liu, T., Yang, J., Yin, Y., Li, M., Wang, L., Zhu, Z., 2026c. Aligning LLM agents with human learning and adjustment behavior: A dual agent approach. *Transportation Research Part C: Emerging Technologies* 191, 105818. doi:10.1016/j.trc.2026.105818.
- Louviere, J.J., Hensher, D.A., Swait, J.D., 2000. Stated choice methods: analysis and applications. Cambridge University Press.
- Martín-Baos, J.Á., López-Gómez, J.A., Rodríguez-Benitez, L., Hillel, T., García-Ródenas, R., 2023. A prediction and behavioural analysis of machine learning methods for modelling travel mode choice. *Transportation Research Part C: Emerging Technologies* 156, 104318. doi:10.1016/j.trc.2023.104318.
- McFadden, D., 1974. Conditional logit analysis of qualitative choice behavior, in: *Frontiers in Econometrics*. Academic Press New York, pp. 105–142.

- Mo, B., Xu, H., Ma, R., Cho, J.H., Zhuang, D., Guo, X., Zhao, J., 2026. Large language models for travel behavior prediction. *Transportation Research Interdisciplinary Perspectives* 38, 102124. doi:[10.1016/j.trip.2026.102124](https://doi.org/10.1016/j.trip.2026.102124).
- Nie, T., Sun, J., Ma, W., 2025. Exploring the roles of large language models in reshaping transportation systems: A survey, framework, and roadmap. *Artificial Intelligence for Transportation* 1, 100003. doi:[10.1016/j.ait.2025.100003](https://doi.org/10.1016/j.ait.2025.100003).
- Park, W., Kim, D., Lu, Y., Cho, M., 2019. Relational knowledge distillation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3967–3976. URL: https://openaccess.thecvf.com/content_CVPR_2019/html/Park_Relational_Knowledge_Distillation_CVPR_2019_paper.html.
- Sanh, V., Debut, L., Chaumond, J., Wolf, T., 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.
- Sifringer, B., Lurkin, V., Alahi, A., 2020. Enhancing discrete choice models with representation learning. *Transportation Research Part B: Methodological* 140, 236–261. doi:[10.1016/j.trb.2020.08.006](https://doi.org/10.1016/j.trb.2020.08.006).
- Train, K.E., 2009. *Discrete choice methods with simulation*. Cambridge University Press.
- Walker, J., Ben-Akiva, M., 2002. Generalized random utility model. *Mathematical Social Sciences* 43, 303–343.
- Wang, L., Duan, P., He, Z., Lyu, C., Chen, X., Zheng, N., Yao, L., Ma, Z., 2025. Agentic large language models for day-to-day route choices. *Transportation Research Part C: Emerging Technologies* 180, 105307. doi:[10.1016/j.trc.2025.105307](https://doi.org/10.1016/j.trc.2025.105307).
- Wang, S., Wang, Q., Zhao, J., 2020. Deep neural networks for choice analysis: Extracting complete economic information for interpretation. *Transportation Research Part C: Emerging Technologies* 118, 102701.
- Xu, Y., Jiao, J., 2027. Benchmarking retrieval-augmented generation strategies for large language model-based travel mode choice prediction. *Travel Behaviour and Society* 46, 101375. doi:[10.1016/j.tbs.2026.101375](https://doi.org/10.1016/j.tbs.2026.101375).
- Yan, Y., Liu, T., Yin, Y., 2026. Valuing time in silicon: Can large language models replicate human value of travel time. *Travel Behaviour and Society* 44, 101245. doi:[10.1016/j.tbs.2026.101245](https://doi.org/10.1016/j.tbs.2026.101245).